

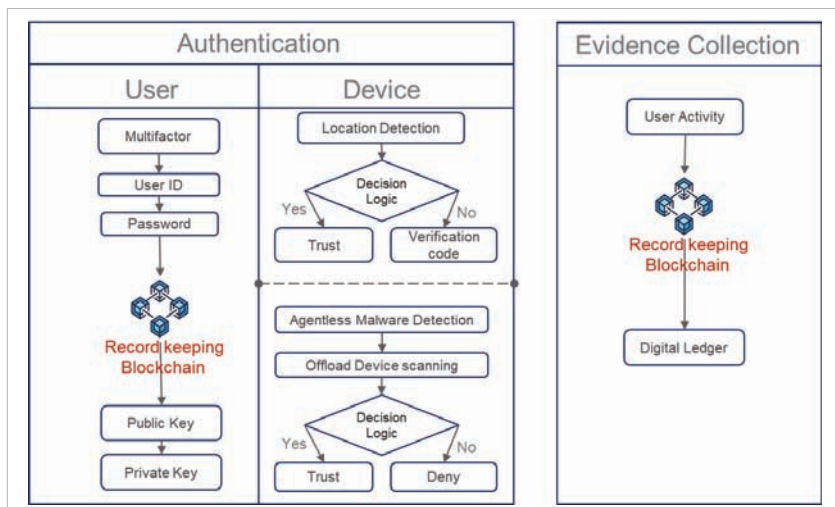


Figure 3: BDP building blocks

Blockchain technology is an immutable time-stamped series of records that is cryptographically secure, distributed and managed by a cluster of nodes. The three pillars of blockchain technology – decentralisation, transparency and immutability – deliver a scalable, resilient and robust backend system along with a new breed of tools for digital identity, authentication and authorisation of users as well as connected devices.

Challenges of BYOD and a blockchain based solution

BYOD is a practice followed by many organisations where employees can use their personal devices for work purposes. It may bring a lot of advantages, but also leads to security issues. Data can be leaked because of the weak authentication technique used to verify the user and the device. A secure authentication technique is what the organisation needs. Blockchain is the answer to this because it uses cryptographic technology that's not easy for a hacker to break. In order to secure sensitive data, a record-keeping model is built using blockchain technology where every activity related to a record is documented in the digital ledger. This ledger can be used as evidence collection for further investigation. This

solution helps organisations to minimise the cases of data leakage while allowing employees to bring their device.

Blockchain is a cryptographic technology that records all user transactions in a digital ledger that is distributed across the network. Blockchain platform is not dependent on any individual entity because the ledger is shared in a decentralised way. It prevents occurrence of human errors, which make it reliable. The record can be accessed from anywhere and the confidentiality of the data is maintained by encryption and hashing methods. The use of cryptography prevents unauthorised access to the network and ensures only legitimate users are allowed to participate.

The key layers and components of a typical blockchain defined perimeter (BDP) are depicted in Figure 3. The security framework consists of authentication and authorisation modules. The authentication process is comprehensive, covering user and device checks. Only authorised users are allowed to access enterprise resources. Enterprises may choose to deploy infrastructure in their data centre, on a public cloud or use a hybrid model. End users may use enterprise managed devices or unmanaged BYOD devices to connect via any kind of network channel to access enterprise resources.

Blockchain technology strengthens zero trust network access in the following ways.

- **Access protection:** This is a form of access management that is designed to let trusted traffic in. A combination of setting access security policies, monitoring usage, and managing usage creates an adaptive access protection model. The authentication trust model comprises user and device authentication with location detection capability. This process prevents data leakage through unauthorised access and malware infection. The first part is a user authentication process. It uses multifactor authentication, where the user needs to key in their ID, password and second factor pass code. The authentication process then goes through the blockchain process, where a private key and public key is assigned to the user. Blockchain uses an asymmetric cryptography mechanism to authenticate transactions. Data on employee private keys is not kept in the database but only stored in the blockchain. Users get a public key during registration, which acts as user identification in the blockchain. The key is kept in the authentication database. The second part is the device authentication process where the device goes through two phases. The first phase is where the malware database server does offload scanning to find out if the device is clean. If this is so, the device goes through the second phase of the authentication, which is location detection. Company policies are applied based on location, and access is granted if conditions are satisfied. As an additional security measure, users are asked to provide a verification code. This authentication is agentless as offload scanning is done through the network without